5(a)	sum of e.m.f.(s) = sum of p.d.(s) **or** (algebraic) sum of e.m.f.(s) and p.d.(s) is zero	**M1**
	around a loop / around a <u>closed</u> circuit	**A1**
5(b)(i)	$1 / r_{(T)} = 1 / 0.59 + 1 / 0.59 + 1 / 0.59$	**B1**
	($r_{(T)}$ =) 0.197 (Ω) (R =) 2.2 – 0.197 = 2.0 Ω	**A1**
	or	
	I = 1.5 / 2.2 (= 0.68 A) **and** i = 0.68 / 3 (where I is the circuit current and i is the current from each cell)	**(B1)**
	($E = IR + ir$ =) 1.5 = 0.68R + (0.68 / 3) × 0.59 **and** R = 2.0 Ω	**(A1)**

Question	Answer	Marks
5(b)(ii)	current = 1.5 / 2.2	**C1**
	= 0.68 A	**A1**
	or	
	p.d. across cell = p.d. across conductor 1.5 – 0.59I = 3I × 2.0 so I = 0.228 A (where I is current in cell) current = 3 × 0.228	**(C1)**
	= 0.68 A	**(A1)**
	or	
	current in conductor = 3 × current in cell V / 2.0 = 3 × (1.5 – V) / 0.59 (where V is p.d. across conductor) V = 1.37 V current = 1.37 / 2.0	**(C1)**
	= 0.68 A	**(A1)**
5(c)(i)	$R = \rho L / A$	**C1**
	$R = 4\rho L / \pi d^2$ (ρ and L are the same so) $R_A / R_B = 7.6^2 / 4.3^2$	**C1**
	= 3.1	**A1**

Question	Answer	Marks
5(c)(ii)	$I = Anvq$ and I, n, q are same / equal / constant	**B1**
	$\frac{v_A}{v_B} = \frac{A_B}{A_A} = \frac{d_B{}^2}{d_A{}^2}$ ratio = $7.6^2 / 4.3^2$ = 3.1	**A1**
5(d)	combined internal resistance (of the cells) will be greater **or** total / circuit resistance (of circuit) greater (because a parallel resistance removed)	**B1**
	more 'lost volts' (inside each cell) **or** internal resistances take a greater share of total p.d. **or** conductor gets a smaller share of the total p.d. **or** current in <u>conductor</u>/total current decreases	**M1**
	(so) potential difference (across conductor) decreases	**A1**